Abubakar Ahmed

Software Engineer/ AI Specialist

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PROFILE

Enthusiastic Software Engineering undergraduate specializing in Machine Learning and Robotics. Passionate about applying technical expertise to real-world challenges, with a strong foundation in software development, Al models, and automation. Experienced in leading training programs, implementing machine learning models, and driving impactful projects. Also deeply passionate about motorsports, leveraging data-driven approaches to enhance race strategies and performance analytics.

EDUCATION

B.Sc. Software Engineering, *African Leadership University* 2023 – 2026 | Kigali, Rwanda

PROFESSIONAL EXPERIENCE

Alpha Rides, *Software Developer* 12/2025 – Present | Remote
- Designed and developed a scalable Flutter-based ride-hailing driver application using modular architecture, Riverpod state management, and GoRouter navigation.
- Implemented real-time ride management features including GPS tracking, Google Maps integration, WebSocket live updates, and push notifications.
- Built secure and production-ready backend communication using Dio, token refresh mechanisms, encrypted storage, and multi-environment deployment for Android and iOS.

CodeFactory, *Facilitator* 02/2025 – 11/2025 | Kigali,Rwanda
- Facilitated Python programming classes for beginners and intermediate learners.
- Supported learners with different skill levels in understanding programming logic and concepts using interactive and project-based methods.
- Provided one-on-one mentorship and technical support to learners and fostered a collaborative and engaging learning environment.

Coderina Edtech, *Trainer* 2023 – 2024 | Kigali, Rwanda
- Taught students how to build, program, and operate robots, using platforms like LEGO Spike, or Arduino.
- Worked with fellow educators, team members, and industry professionals to stay updated on the latest trends in robotics.
- Trained the students and Hosting the annual First Lego League progam.

Ikra Foundation for Women and Youth Development, *VOLUNTEER/ IT Specialist* 2020 – 2022 | Bauchi, Nigeria
- Assisted with troubleshooting hardware/software issues, setting up devices, and helping employees with IT-related concerns.
- Volunteered in consolidating advocacy for justice delivery to survivors of Sexual and Gender-Based Violence (SGBV) in Nigeria.
- Supported effective legal frameworks and survivor support systems Assisted in the publication and dissemination of the Violence Against Persons Prohibition (VAPP) Law and the Administration of Criminal Justice Laws across all 20 LGAs of Bauchi State.

PROJECTS

EcoDrive Simulator
- Developed an interactive web-based simulator using reinforcement learning (PPO & SAC) to optimize eco-driving trajectories and strategies on Formula 1 tracks.
- Built the system end-to-end, including real-time telemetry, model explainability, and a FastAPI backend.

Tire-Wear Prediction App
- Built a machine learning application to predict tire wear percentage based on driving conditions using a Scikit-Learn regression model (85% accuracy).
- Deployed the model via FastAPI and integrated it into a Flutter mobile application.

Race Strategy Optimizer
- Developed a machine learning model to optimize Formula 1 tire selection using 100,000+ data points on track conditions, weather, and driving style.
- Improved prediction accuracy by 15% using Logistic Regression and Neural Networks

SKILLS

Technical Skills: Programming Languages: Python, JavaScript, Dart (Flutter) | Machine Learning & Al: Scikit-Learn, TensorFlow, Neural Networks | Web & Mobile Development: HTML5, CSS, JavaScript, Flutter | Databases: MySQL, Firebase | Cloud & Tools: Google Colab, FastAPI, Git/GitHub, AWS

Soft Skills: Project Management | Public Speaking & Advocacy | Teamwork | Time Management | Leadership | Effective Communication | Critical Thinking

CERTIFICATES

- Exploratory Data Analysis for Machine Learning - IBM
- Structuring Machine Learning Projects - DeepLearning.AI
- Python Data Structures - Universtity of Michigan
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization - DeepLearning.AI
- Getting Started with R - Coursera
- Python Programming for Everybody - University of Michigan
- Neural Networks and Deep Learning - DeepLearning.AI